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Prof. Dr. Anna Poetsch
Poetsch Group — Institut für Genetik / CECAD
Universität zu Köln
apoetsch@uni-koeln.de
Ref. Wiss2604-09

Re: Deep Learning Engineer — Omics Data / Genomics (Ref. Wiss2604-09)

Dear Prof. Poetsch,

I am applying for the Deep Learning Engineer position in the Poetsch group (Ref. Wiss2604-09). The position describes building deep learning infrastructure on omics data for the lab and the broader research community — converting theoretical concepts into working code, implementing state-of-the-art algorithms, and creating internal tools that accelerate research. This is exactly the kind of work I have been doing independently for four years, across genomics, transcriptomics, mass spectrometry, and immunopeptidology data. The frameworks I have built are not academic exercises; they are deployed, validated, and operational systems that I can bring into the lab's infrastructure.

Deep learning for genomics and omics data. The Gospel framework implements the full genomics analysis pipeline at production scale: 10^6 variants per second VCF processing, RNA-seq quantification and differential expression, Bayesian network inference for multi-omics integration, and Gaussian mixture models for population structure. Validated against 1000 Genomes Phase 3, gnomAD v3.1.2, and UK Biobank. For the Poetsch group's work on genomic changes in aging and cancer, Gospel provides the computational backbone for somatic variant calling, tumour mutational burden analysis, and integrating genomic with transcriptomic data across patient cohorts.

The Lavoisier framework demonstrates the specific DL pattern the position describes — implementing and evaluating a state-of-the-art algorithm from a recent academic paper and deploying it as a research tool. The dual-pipeline design combines:

- A numerical pipeline (standard MS1/MS2 processing, peak detection, spectral library matching) validated against NIST at 98.9% accuracy.
- A CNN-based visual pipeline (PyTorch) that converts mass spectra into temporal image sequences and applies image classification networks — moving from the academic literature on spectral imaging into a working, distributed production system (Ray, >100 GB mzML support).

This is precisely the "from academic papers to working code" task the position lists as a core responsibility. The same architectural pattern — parallel classical and deep

learning pipelines with cross-validated accuracy metrics — applies directly to genomics data: a classical variant calling pipeline alongside a CNN or Transformer-based variant pathogenicity predictor, both sharing the same input pre-processing layer.

Algorithm implementation and internal tooling. The position emphasises creating internal tools that speed up research: automated evaluation frameworks, data annotation tools, specialized libraries. I have built all three categories:

Automated evaluation frameworks: Every framework I have built includes automated validation against reference standards with quantified accuracy metrics. The Lavoisier pipeline runs automated benchmarking against the NIST mass spectral library on each build; Gospel validates variant call accuracy against gnomAD population frequencies and Clinvar annotations; the Syndrome immunopeptidology tools benchmark against the IEDB peptide database.

Data annotation tools: The Syndrome live tools provide browser-based annotation for genomic sequences — MHC binding prediction, neoantigen identification, immune escape scoring — requiring no local installation. This is the kind of interactive annotation tool that lab members with varying computational expertise can use directly.

Specialised libraries: The Rust core of Four-Sided-Triangle is a specialised library for high-throughput Bayesian inference, fuzzy logic, and evidence propagation, providing 15–50× speedup over Python via PyO3 bindings. The same library architecture — Rust performance-critical operations exposed to Python via clean API — is the standard approach for building genomics DL libraries where inference throughput matters.

Aging and cancer genomics: domain context. The Poetsch group’s focus on genomic changes in aging and cancer is the application domain I have studied most intensively from a computational perspective. For cancer: the Syndrome framework’s four live immunopeptidology tools (neoantigen identification, MHC binding, immune escape, drug resistance) directly analyse the somatic mutations and immune interactions relevant to cancer genomics. The Crown Prince platform derives tumour cell diagnostics from spectral hologram representations, with $AUC = 1.00$ on the loop-holonomy self-consistency diagnostic. For aging: the Olduvai framework analyses biological motion degradation patterns across aging (Parkinson’s, ataxia, healthy ageing), validated on 86 nights of wearable sensor data. The $D \in [0, 1]^8$ disease vector in Syndrome has an explicit circadian regulatory oscillator dimension (D_{Circ}) and ATP synthesis coherence dimension (D_{ATP}) — the two cellular dimensions most directly coupled to the aging process at the genomic expression level. Connecting genomic variation data (from Gospel’s variant analysis) to these cellular phenotype dimensions is the kind of integrative analysis the position is structured to enable.

Equivalent experience in place of a completed doctorate. The position specifies a PhD in CS, Bioinformatics, or AI “or equivalent experience level.” I completed doctoral research in Computational Lipidomics at TUM / Bayerisches Forschungsinstitut (2019–2021) and have since worked full-time as an independent researcher, producing peer-reviewed-quality manuscripts (three submitted, affiliated with AIME Registry and TUM Weihenstephan) and building twelve deployed, validated computational platforms. The depth and scope of this independent research is the equivalent experience the position’s clause refers to.

I work primarily in Python with PyTorch; Rust for performance-critical components; Ray

for distributed computing; Docker and Kubernetes for deployment. English is native; German is conversational. I am legally resident in Germany with full work authorisation and available from July 2026.

Yours sincerely,

Kundai Farai Sachikonye